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Stat 143 – TTh 2:30pm-4:00pm **28 February 2017**
Revised Proposal Paper

I) Title

HYGIENE PRACTICES OF FOOD HANDLERS ON DIFFERENT FOOD ESTABLISHMENTS IN TEACHER'S VILLAGE AREA

II) Background of the study

Teacher's Village is slowly making a name for featuring scrumptious dining experience. It has grown to become the go-to place for food seeking individuals on the lookout for the best restaurants serving delectable, savory dishes.

Increasingly popular go-to strip has evolved into a food paradise. Walk this street and you'll find yourself enticed by the variety of food offerings ranges from Western, exotic, Eastern, and Pinoy local cuisines. And this leads to an increase in the establishment of many eating places such as coffee shops, bars, bakeries, Tea shops and restaurants.

Food is an essential part of everyday life and managing its preparation is important to make sure the food you serve is safe to eat. People handling food should observe certain strict hygienic measures when it comes to cleaning, preserving, cooking or storing food for human consumption. This is because good health is reliant on the manner food is handled.

The safety of consumers is the paramount responsibility of any food service establishment. Food handlers are thus expected to observe proper hygiene and sanitation methods as the chances of food contamination largely depend on their health status and hygiene practices (O Ifeadike, C Ironkwe, O. U. Adogu, & C Nnebue, 2014).

In this line of thought, this study will be conducted to evaluate the food hygiene knowledge and practices of food handlers of food service establishments in Teacher's Village. The output of this study may serve as a tool for information dissemination.

III) Review of Related Literature

Effective hygiene control is vital to avoid the adverse effects to human health and economic consequences of foodborne illness, foodborne injury, and food spoilage. Everyone, including farmers and growers, manufacturers and processors, food handlers and consumers have a responsibility to assure that food is safe and suitable for consumption (FAO/WHO, 2005)

Food handler is defined as a person in food trade or someone professionally associated with it, such as an inspector, who in his routine work comes into direct contact with food in the course of production, processing, packaging or distribution. (R. Kubde, Pattankar, & R. Kokiwar, n.d.)

U.S. consumers stress the need for knowledgeable, well-trained food service and retail food stores employees have changed food preference with many people/consumers away from home now eating raw foods or slightly cooked foods that can increase the risk of foodborne illness. (Marriott, Gravani 2006, pp. 7)

The skin, fingers, fingernails, jewelry, hair, eyes, mouth, and nose are just some of the parts of the human bodies could be potential sources of bacterial contamination. (Marriott, Gravani 2006)

Leocadio (1998) mentioned in her study that the amount of bacteria is influenced by unhygienic preparation. She suggested that in order to ensure sanitation, utensils must be washed and cleaned before using. Storing does not only apply to foods but also to utensils used in the operation. These may also be the source of microorganism that contaminates food and eventually lead to physical, chemical or biological hazards

Poor personal hygiene frequently contributes to foodborne illness which indicates that food handlers' knowledge and handling practices needs to be improved. A study conducted among food handlers in Bahir Dar town indicated that most of them were infected with enteric bacteria and parasites (Abera, Biadegelgen, & Bezabih, 2010). Good personal hygiene and food handling practices are the basis for preventing the transmission of pathogens from food handlers to the consumers.

Food sanitation prevents food poisoning outbreaks, most of the food related illness caused by street food outlets, restaurants, chain restaurants are the result of unsanitary food handling practices. Just one case of food poisoning, is enough to destroy your business. Hence, training employees to apply and follow safe food handling practices is a requirement (Q. GADUENA, n.d.). Sanitation is more than cleanliness. If done properly, it can improve the aesthetic qualities and hygienic conditions of commercial operations. Given if customers do not get sick, preparing or serving food in unsanitary condition will surely affect food quality and taste. Keeping things properly and sanitary practices will no longer look into your food preparation area.

Chipabika's (2013) study revealed that there was an association between level of knowledge and hygiene practices among food handlers. Furthermore, the study revealed that food handlers that were very knowledgeable fell among those whose level of education was secondary school. And within this category, those without any form of education were the least knowledgeable.

Republic Act No. 10611, "An act to strengthen the food safety regulatory system in the country to protect consumer health and facilitate market access of local foods and food products, and for other purposes" otherwise known as the "Food Safety Act of 2013", where the Department of Agriculture (DA) and the Department of Health (DOH) jointly adapt and promulgate rules and regulations for food safety in the Philippines. ("Official Gazette," 2013)

Similar studies should be undertaken in order to monitor the quality of food being offered to the public and also protect the health of the people and also the economy of the country because it has negative effects on the economy of the country.

Evaluation during continuous supervision and self-inspection should be conducted for effectiveness. Monitoring is also necessary to verify that the procedures are followed. And it should be documented in the form of periodic inspection reports to verify that the sanitation program is being followed and that expected results have occurred. (Marriott, Gravani 2006)

IV) Statement of the Problem

Food is highly perishable commodity which can directly affect the health of a consumer. To ensure availability of safe and hygienic food to consumers is challenge for food service establishments and regulatory authorities.

This study will focus on the evaluation of Food Hygiene Practices of Food Handlers on different Food Establishments in Teacher's Village Area, and will specifically pursue the following objectives:

a) To establish the characteristics of food handlers working in randomly selected restaurants in Teacher's Village

b) To assess the level of knowledge regarding food hygiene practices among food handlers in Teacher's Village

c) To evaluate hygiene practices of food handlers on different food establishments in Teacher's village

IV) Definition of variables and how they will be measured

In this research study, the dependent variable is the food hygiene practices of food handlers in the different food establishments in barangay Teacher's Village while the independent variables are the characteristics of food handlers, levels of knowledge and practices on food hygiene of the food handlers in selected food establishment in Teacher's Village.

VARIABLE NAME	DESCRIPTION	MEASUREMENT
Dependent Variable		
FOOD HYGIENE PRACTICES	Acceptable Standards – 1. Examination certificates for food handlers, 2. Appropriate dress code 3. Running water with soap 4. Receptacle for waste storage 5. Hand washing with soap 6. Paper towel	Ordinal; • Very good = 6 (81%- 100%) • Good = 4-5 (60-80%) • Fair = 2-3 (50%-59%) • Poor = below 2 (49%
Independent Variable		
SEX	Male or Female	Nominal
AGE	Number of years at last birthday	Ratio; - 15-20 years old - 21-26 years old, - Above 26 years old
EDUCATIONAL LEVEL	Highest Educational Attainment	Ratio/Ordinal - Primary and Below, - Secondary, - Tertiary
TRAININGS	Training of Food handlers in food hygiene practices	-Trained -Not trained
DEFINITION	Definition of food hygiene for food handlers	- It is the action taken to ensure that food is handled, stored, prepared, and served in such a way to prevent the contamination of food.

		-It is the way of maintaining personal hygiene. -It is maintaining cleanliness of the environment.
NO OF TIMES FOR HANDWASHING	No of times when hands are washed in the food establishments	-Two Times -Three Times -Every after procedure
KNOWLEDGE	Level of Knowledge: 1. Posting of food hygiene information sheet 2. Importance of food hygiene training 3. Cross Contamination 4. Prevention of foodborne diseases 5. Good hygiene practices prevents diarrhea	Ordinal; • Very knowledgeable = >80% • Knowledgeable = >60 - 79% • Moderate = >50 - 59% • Poor = < 50%
PRACTICES	Hygiene Practices of Food Handlers: 1. Wash hands before handling food 2. No of times when hands are washed (Once a day, Three times a day, After every procedure) 3. Wash hands with soap 4. Covering hair 5. Protective clothing 6. Washing Cooking utensils in hot water 7. No of times when nails are cut (Once a week, Twice a month, Once a month) 8. No of times when working area is cleaned (one time, two times, every after a procedure)	Ordinal; • Very knowledgeable = >80% • Knowledgeable = >60 - 79% • Moderate = >50 - 59% • Poor = < 50%
LEGAL ENFORCEMENT	Number of Inspections done in last quarter	Ordinal; • Very knowledgeable = >80% • Knowledgeable = >60 - 70% • Fair = >50 - 59% • Poor = <50%
WASTE MANAGEMENT	Amount of waste generated	Present or Absent

VI) Schematic Diagram

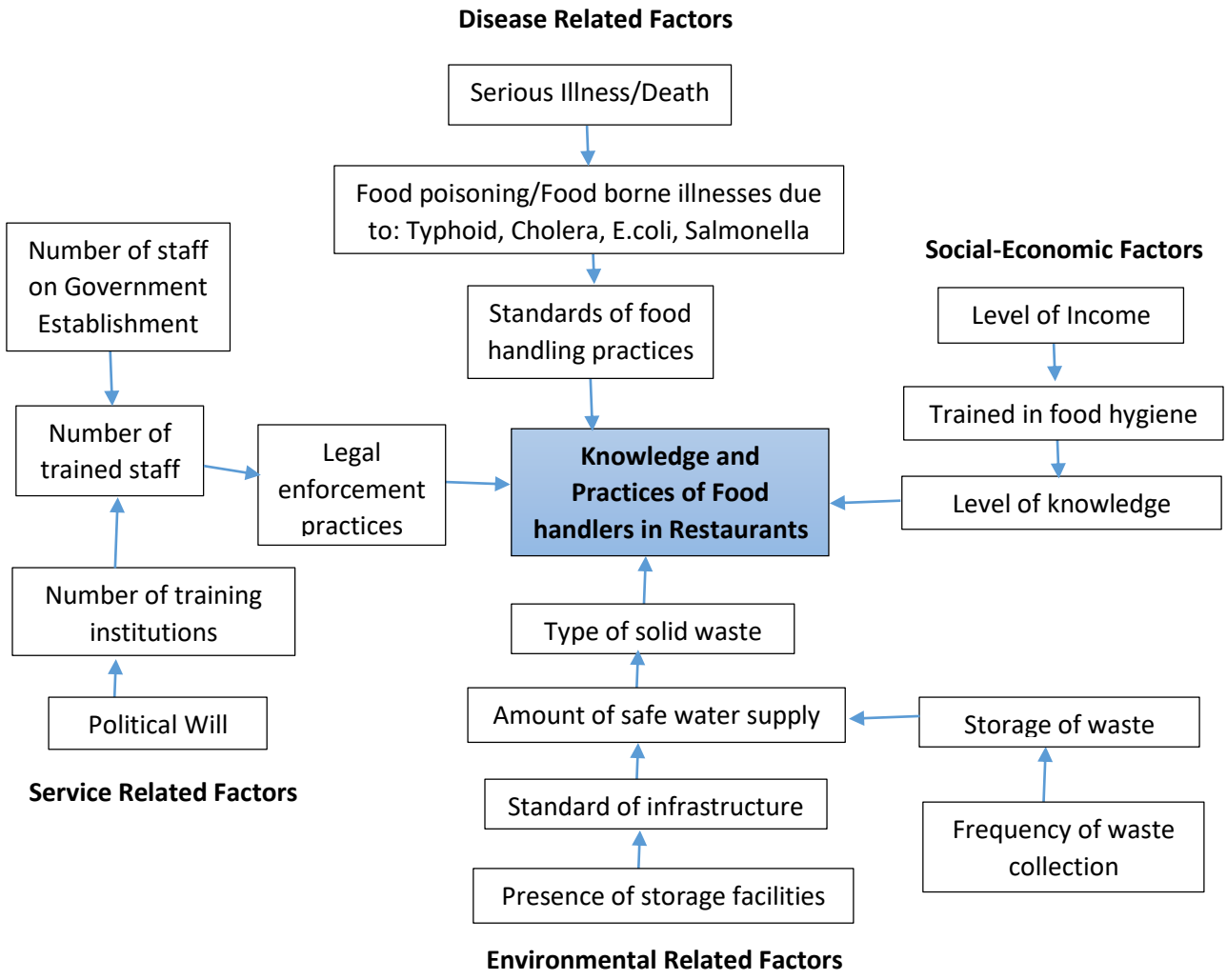


Figure 1. Problem Analysis Diagramme. Adapted from "An Assessment of Food Hygiene Practices among food handlers in restaurants in Kabwe Urban District," by Ennie Chipabika, 2013, p. 12.

Food handlers can carry microbial pathogens on their skin, hair, hands, digestive systems or respiratory tracts and unless they have a thorough understanding and follow basic food hygiene principles, they may unintentionally contaminate foods, water supplies and equipment thereby creating favorable conditions for an outbreak of foodborne illnesses (Dugassa, 2007).

Water is a critical raw material in food establishments which may equally be contaminated with biological, chemical or physical hazards. As such, contaminated water will create a public health risk if it is used for drinking purpose, washing of food, incorporated into food as an ingredient and used in food processing, washing of equipment, utensils and containers (Sprenger, 2006). Waste disposal if not properly implemented may lead to food contamination.

VII) Limitations of the study

This study only deals with the food hygiene practices of food establishments in the Barangay Teacher's Village. Specifically, this study will be conducted to establish the characteristics of food handlers working in randomly selected restaurants, to assess the level of knowledge regarding food hygiene practices among food handlers and to evaluate hygiene practices of food handlers on different food establishments in Teacher's village. This is only limited to all the food handlers who are working on the food establishments and are willing to participate in the research study.

Thus, the results will serve as a guide for the people to have at least knowledge on how the food handlers in Barangay Teacher's village prepare their food and in other food establishments here in our country.

VIII) Statistical Tools to be used

a) Hypothesis to be Tested – specific hypothesis tests to be conducted and associated dummy tables

H_o: There is no relationship between the level of knowledge and food hygiene practices of food handlers in the food establishments in Barangay Teacher's Village.
H_a: There is a relationship between the level of knowledge and food hygiene practices of food handlers in the food establishments in Barangay Teacher's Village.

b) Method of Data Collection including computation of Sample Size – stat model and associated dummy tables

The study population is the 142 food establishments in the Teacher's Village Area and 1000 food handlers. The food handlers consist of female and male personnel who work in identified food establishments.

This study will draw respondents or sample units using simple random sampling method from the sampling frame. This will be done by assigning numbers to food handlers in each food establishments and all the numbers will put in the box. Then each unit will be picked randomly from the box. Three (3) food handlers will be randomly picked from each restaurants. Simple random sample will be used to obtain a representative sample and get information which will be the reality on the ground.

Taking 142 as the total number of food establishments and 1000 as the sampling frame for food handlers, the sample size is calculated as follows:

$$n_0 = \frac{z^2{}_{\alpha/2}\hat{p}(1-\hat{p})}{e^2}$$

where, n_o = sample size without considering the finite population correlation factor, z = standard normal deviate at 95%, confidence interval = 1.96, $\hat{p}$ = proportion of food handlers with acceptable food hygiene practices – 50% (A,20**), e = 5% margin of error.

Therefore: $n_0 = \frac{z^2{}_{\alpha/2}\hat{p}(1-\hat{p})}{d^2} = \frac{(1.96)^2*(0.50)*(1-0.50)}{(0.05)^2} = 384.16$

Using finite population correction with N = 1000, leads to $n = \frac{n_0 N}{n_0 + (N-1)} = \frac{384.16(1000)}{384.16 + (1000-1)} = 278$
Thus, the study is set to have 278 participants.

c) Statistical procedure to be used – specific descriptive stat to be computed and associated dummy tables

Findings in this study may be presented through frequency tables, charts, cross tabulations and logistic regression model.

Table 1: Background Characteristics of Food Handlers (n=278)

Demographic Profile	Frequency/Percentage
Sex	
• Male • Female	
Age in Years	
• 15-20 years • 21-26 years • Above 26 years	
Educational Level	
• Primary and Below • Secondary • Tertiary	

Table 2: Knowledge of food handlers on Food Hygiene Practices

	Frequency	Percentage
Posting of food Hygiene Information Sheet		
Posting of food hygiene information sheet No Posting of food hygiene information sheet		
Importance of food Hygiene Training		
Important Not Important		
Cross Contamination		
Poor hygiene can cause cross contamination Poor hygiene cannot cause cross contamination		
Prevention of foodborne diseases		
Aware Not aware		
Good hygiene practices Prevent diarrhea		
Prevent Does not prevent		

Table 3: Hygiene Practices of Food handlers (n=278)

	Frequency	Percentage
Wash hands before handling food		
Wash hands before handling the food		
Do not wash hands before handling food		
No of times when hands are washed		
Once a day		
Three times a day		
After every procedure		
Wash hands with soap		
Wash with soap		
Does not wash with soap		
Covering hair		
Covered		
Not covered		
Protective clothing		
Wear protective clothing		
Do not wear protective clothing		
Washing of cooking utensils in hot water		
Do not wash cooking utensils in hot		
Wash cooking utensils in hot water		
No of times when nails are cut		
Once a week		
Twice a month		
Once a month		
No of times when working area is cleaned		
One time		
Two times		
Every after a procedure		

Table 4: Association between the Education Level and Practices of Food Handlers (n=278)

PRACTICES OF FOOD HANDLERS n (%)					
EDUCATION	**Very Good**	**Good**	**Fair**	**Poor**	**Total**
None					
Primary					
Secondary					
College					
TOTAL					278 (100%)

Table 5: Association between the Education Level and Level of Knowledge on Food Hygiene (n=278)

LEVEL OF KNOWLEDGE ON FOOD HYGIENE n (%)					
EDUCATION	**Very Knowledgeable**	**Knowledgeable**	**Moderate**	**Poor**	**Total**
None					
Primary					
Secondary					
College					
TOTAL					278 (100%)

Table 6: Association between the Level of Knowledge on Food Hygiene and Food Hygiene Practices (n=278)

LEVEL OF KNOWLEDGE ON FOOD HYGIENE n (%)					
EDUCATION	**Very Good**	**Good**	**Fair**	**Poor**	**Total**
Very Knowledgeable					
Knowledgeable					
Moderate					
Poor					
TOTAL					278 (100%)

For tables' 4, 5 and 6, in order to determine whether there is an association between the level of knowledge and hygiene practices, Pearson's Chi-square test will be used. However, we will perform Fisher's Test if there will be less than five (5) expected frequency in some cells, which will make the results in Pearson's Chi-square to be invalid.

Table 7: Logistic Regression Model factors contribute to food hygiene practices of different food establishments (n=278)

Hand washing with soap	Odds Ratio	P>z	95% Confidence Interval	
Demographics				
sex (1=male,0=female)				
Age Group				
18 to 20 years				
21 to 26 years				
Above 26 yearrs				
Educational Level				
Primary and Below				
Secondary				
Tertiary				
Knowledge On Food Hygiene				
Trained in Food Hygiene				
Presence of a Food hygiene statement policy				
Food hygiene training is important				
Knowledge of good hygiene practices				
Food hygiene prevent diarrhea diseases				
Does poor personal hygiene contribute to food contamination				
Definition of food hygiene				
Definition of food-borne diseases				
Constant				

Table 7 will present the results of the logistic regression model which will show whether the level of education has a significant effect on the washing hands with soap which is a good hygiene practice.

IX) Data Collection Tool

Data in this study will be collected using a structured questionnaire which will administer to food handlers. The designed questionnaire focuses on the food handlers' demographic profile, knowledge and practices regarding the use of preventive measures.

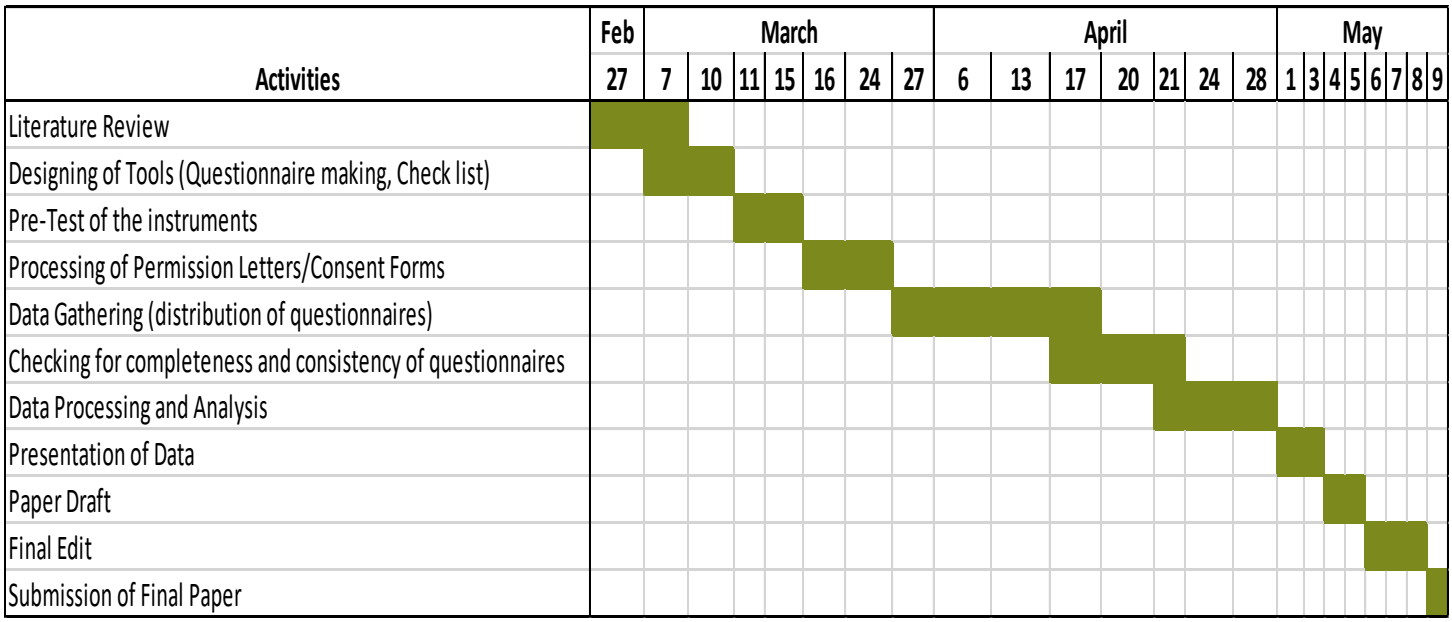
A check list will also be provided to observe behaviors of food handlers in terms of hand washing, washing utensils in hot water, licking fingers and covering hair. It will be also used to assess the sanitary facilities and waste management in the restaurants.

The data collection tools will be pre-tested on food handlers in the food establishments in Teacher's village area. This will be done in order to determine the quality of the information which will be collected using the mentioned tools.

X) Proposed Budget

Budget Allocation	Estimated Cost
Pre-Test of the Instruments	
Fare	Php 1050.00
Printing of draft questionnaires	Php 200.00
Total	**Php 1250.00**
Processing of Permission Letter/Consent Forms	
Printing of needed documents	Php 500.00
Fare	Php 2000.00
Total	**Php 2500.00**
Data Gathering/Collection	
Interviewers' Allowance	Php 45000.00
Fare	Php 7000.00
Printing of Questionnaires and Checklist	Php 3500.00
Total	**Php 55500.00**
Data Processing and Analysis	
Data Analytics	Php 10000.00
Discussion room rentals	Php 3000.00
Internet	Php 500.00
Total	**Php 13500.00**
Printing of Draft Paper	**Php 500.00**
Printing and Binding of the Final Paper	**Php 1000.00**
Grand Total	**Php 74,250.00**

XI) Schedule of Activities



XII) List of References

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Survey Number _____

UNIVERSITY OF THE PHILIPPINES DILIMAN
SCHOOL OF STATISTICS
Survey-Questionnaire

HYGIENE PRACTICES OF FOOD HANDLERS ON DIFFERENT FOOD ESTABLISHMENTS IN TEACHER'S VILLAGE AREA

Name of Food Establishment: Date:

Instruction: Please read carefully and check the corresponding boxes for your answer(s).

Section A: Demographic Data

1. Sex of the Respondent:
   - ☐ Male
   - ☐ Female
2. Age (number of years) at last birthday
   - ☐ 18-20 years
   - ☐ 21-26 years
   - ☐ Above 26 years
3. Highest Education Attainment
   - ☐ None
   - ☐ Primary
   - ☐ Secondary
   - ☐ College

Section B: Knowledge on Food Hygiene

4. Are you trained in food hygiene?
   - ☐ Yes
   - ☐ No
5. Is there a food hygiene statement policy in this food establishment?
   - ☐ Yes
   - ☐ No
6. Is training in food hygiene important in the food industry?
   - ☐ Yes
   - ☐ No
7. Do you know any good food hygiene practices?
   - ☐ Yes
   - ☐ No
8. Mention some of the good hygienic practices you know

   _____
   _____
   _____

9. Are you aware of your responsibilities regarding food hygiene?
   - ☐ Yes
   - ☐ No

10. Can food hygiene prevent diarrhea diseases?
    - ☐ Yes
    - ☐ No
11. Does poor personal hygiene contribute to food contamination?
    - ☐ Yes
    - ☐ No
12. What is food hygiene?
    - ☐ It is the action taken to ensure that food is handled, stored, prepared, and served in such a way to prevent the contamination of food.
    - ☐ It is the way of maintaining personal hygiene.
    - ☐ It is maintaining cleanliness of the environment.
13. What are food-borne diseases?
    - ☐ These are diseases transmitted through consumption of contaminated food
    - ☐ Diseases caused by dinking contaminated water
    - ☐ Diseases that are transmitted by mosquitoes
14. Do you know any food-borne diseases?
    - ☐ Yes
    - ☐ No
15. Can food-borne diseases be prevented?
    - ☐ Yes
    - ☐ No
16. Can licking hands contaminate food?
    - ☐ Yes
    - ☐ No
17. Can skin infections contaminate food?
    - ☐ Yes
    - ☐ No
18. When should the hands be washed in the food establishment?
    - ☐ Two times
    - ☐ Three Times
    - ☐ Every after procedure
19. Should the floor, wall, roof be kept clean?
    - ☐ Yes
    - ☐ No

Section C: Practice

20. Do you wash hands before handling food?
    - ☐ Yes
    - ☐ No
21. How often do you wash your hands when handling food?
    - ☐ Once a day
    - ☐ Three times a day
    - ☐ After every procedure
22. Do you wash your hands with soap?
    - ☐ Yes
    - ☐ No

23. Do you keep your hair covered?
    - ☐ Yes
    - ☐ No
24. Cooking utensils should be washed in hot water?
    - ☐ Yes
    - ☐ No
25. Are you provided with personal protective clothing by management when working in this food establishment?
    - ☐ Yes
    - ☐ No
27. What do you do with leftover food?
    - ☐ Serve it the following day
    - ☐ Throw it away
28 How often do you cut your nails
    - ☐ Once a week
    - ☐ Twice a month
    - ☐ Once a month
    - ☐ There's no need.
29. How many times do you clean your working area?
    - ☐ Once
    - ☐ Twice
    - ☐ Every after a procedure
30. What do you use to clean your working area?
    - ☐ Water and soap
    - ☐ Water only
31. What do you use to wash your kitchen utensils in the food establishment?
    - ☐ Water and soap
    - ☐ Water only
32. Do you have running water in this food establishment?
    - ☐ Yes
    - ☐ No
33. Are you provided with change rooms in this food establishment?
    - ☐ Yes
    - ☐ No
34. Are you provided with shower rooms in this food establishment?
    - ☐ Yes
    - ☐ No

Thank you for sparing this time to assist us in our study!

Adapted and modified from the study "An Assessment of Food Hygiene Practices among food handlers in restaurants in Kabwe Urban District," by Ennie Chipabika, 2013, p. 45.

UNIVERSITY OF THE PHILIPPINES DILIMAN
SCHOOL OF STATISTICS
Checklist

HYGIENE PRACTICES OF FOOD HANDLERS ON DIFFERENT FOOD ESTABLISHMENTS IN TEACHER'S VILLAGE AREA

Theme	**Logistics**	**Yes**	**No**	**Remarks**
1. Personnel	Is there evidence of medical examination for food handlers?			
	Is there evidence of training in food safety?			
	Do food handler wear Personal Protective clothing?			
	Cutting of finger nails			
2. Handwashing Facilities	Running water (Hot and Cold)			
	Hand dryer			
	Sink/Basin			
	Soap			
3. Utensils	Are there appropriate storage facilities?			
	Presence of washing facilities			
	Presence of drying facilities			
4. Water Supply	Running water			
	Water			
5. Waste management	Storage facilities			
	Record on amount of waste generated			
	Records on waste collection			
6. Sanitary facilities	Sanitary facilities (male and female)			
	Change rooms			
	Showers for male and females			
	Hand washing facilities			
7. Documentation	Is there a policy for maintaining food safety documents?			
	Documentation on trainings			

Adapted and modified from the study "An Assessment of Food Hygiene Practices among food handlers in restaurants in Kabwe Urban District," by Ennie Chipabika, 2013, p. 57.